**EXPERIMENT 1:**

**Characteristics Study of Side Band Power and Frequencies in AM Using MATLAB**

**Aim:** To generate an Amplitude Modulated (AM) signal using MATLAB and study its sideband frequencies and power. To calculate the carrier power, upper and lower sideband power, and total AM power.

**Soft ware Required:** MATLAB

**Program:**

```
clc;

clear;

close all;% Characteristics Study of Sideband Power and Frequencies in AM

% Step 1: Define parameters

Ac = 10;         % Carrier amplitude (V)

Am = 5;          % Message signal amplitude (V)

fc = 1000;       % Carrier frequency (Hz)

fm = 100;        % Message frequency (Hz)

Fs = 10000;      % Sampling frequency (Hz)

t = 0:1/Fs:0.05;  % Time vector

% Step 2: Calculate modulation index

m = Am / Ac;

fprintf('AMPLITUDE MODULATION ANALYSIS\n');

fprintf('----------------------------\n');

fprintf('Carrier amplitude = %.2f V\n', Ac);

fprintf('Message amplitude = %.2f V\n', Am);
```

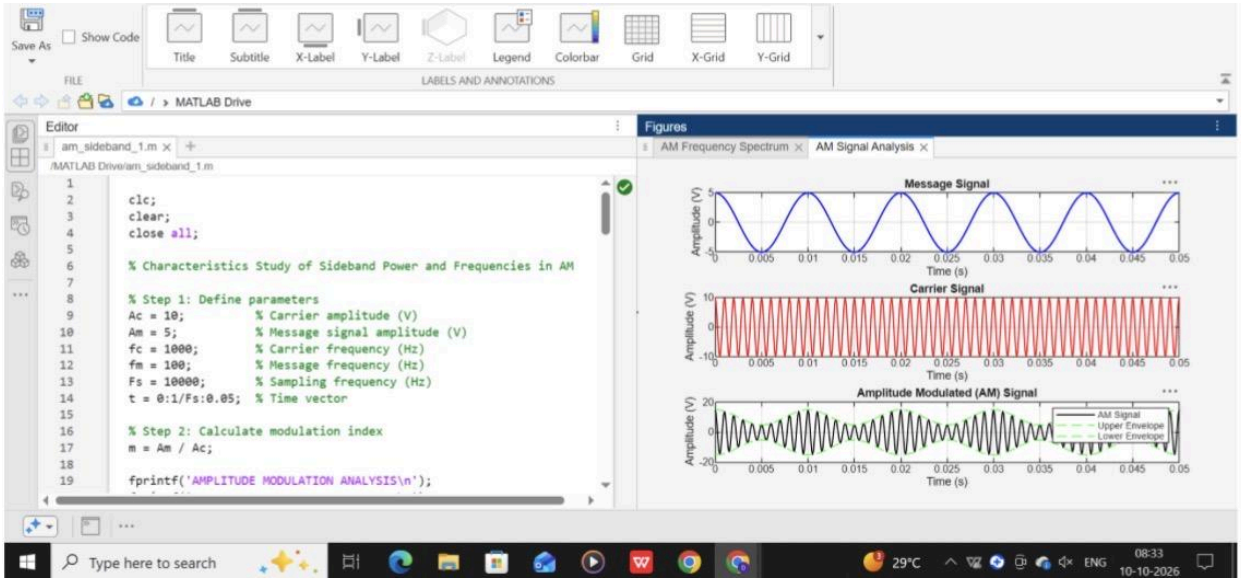
```
fprintf('Carrier frequency = %.2f Hz\n', fc);
fprintf('Message frequency = %.2f Hz\n', fm);
fprintf('Modulation index = %.2f (%.1f%%)\n', m, m*100);
% Step 3: Generate carrier and message signals
carrier = Ac * cos(2*pi*fc*t);
message = Am * cos(2*pi*fm*t);
% Step 4: Generate AM signal
am_signal = Ac * (1 + m*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);
% Step 5: Calculate sideband frequencies
f_upper = fc + fm;
f_lower = fc - fm;
fprintf('\nFREQUENCY ANALYSIS\n');
fprintf('------------------\n');
fprintf('Lower sideband frequency = %.2f Hz\n', f_lower);
fprintf('Carrier frequency = %.2f Hz\n', fc);
fprintf('Upper sideband frequency = %.2f Hz\n', f_upper);
fprintf('Bandwidth = %.2f Hz\n', f_upper-f_lower);
% Step 6: Calculate power (assuming 1-ohm load)
Pc = Ac^2 / 2;              % Carrier power
Psb_each = (m^2 * Ac^2)/8;   % Power in each sideband
Psb_total = 2 * Psb_each;   % Total sideband power
Ptotal = Pc + Psb_total;    % Total AM power
```

```
fprintf('\nPOWER ANALYSIS (1-ohm load)\n');
fprintf('--------------------------\n');
fprintf('Carrier power = %.2f W\n', Pc);
fprintf('Lower sideband power = %.2f W\n', Psb_each);
fprintf('Upper sideband power = %.2f W\n', Psb_each);
fprintf('Total sideband power = %.2f W\n', Psb_total);
fprintf('Total AM power = %.2f W\n', Ptotal);
% Step 7: Plot time-domain signals
figure('Name','AM Signal Analysis','NumberTitle','off');
subplot(3,1,1);
plot(t, message, 'b', 'LineWidth', 1.2);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,2);
plot(t, carrier, 'r', 'LineWidth', 1.0);
grid on;
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,3);
plot(t, am_signal, 'k', 'LineWidth', 1.0);
```

```
hold on;
plot(t, Ac*(1+m*cos(2*pi*fm*t)), 'g--');
plot(t, -Ac*(1+m*cos(2*pi*fm*t)), 'g--');
hold off;
grid on;
title('Amplitude Modulated (AM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
legend('AM Signal','Upper Envelope','Lower Envelope');
% Step 8: Frequency spectrum using FFT
N = length(am_signal);
Y = fft(am_signal);
P2 = abs(Y/N);
P1 = P2(1:floor(N/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
f = Fs*(0:floor(N/2))/N;
figure('Name','AM Frequency Spectrum','NumberTitle','off');
plot(f, P1, 'b', 'LineWidth', 1.3);
grid on;
title('Frequency Spectrum of AM Signal');
xlabel('Frequency (Hz)');
ylabel('Amplitude (V)');
xlim([0 1500]);
```

```
% Step 9: Display theoretical frequency components
hold on;
stem([f_lower fc f_upper], [Ac*m/2 Ac Ac*m/2], ...
   'r', 'filled');
hold off;
legend('FFT Spectrum','Theoretical Components');
```

**Output:**




**Result:** The AM signal was successfully generated and its frequency spectrum was analyzed using MATLAB. The carrier frequency, sideband frequencies, and corresponding power values were calculated and verified.